

Class 9
Computer Applications(2026-27)
Chapter: 5

Operators In Java

(Solutions are below the questions)

Unsolved Questions

✓ A. Tick (✓) the correct answer

1. What values will be stored in x and y respectively?

```
int x = 10, y; y = -x;
```

- a. x = -11 and y = -11
- b. x = 10 and y = -10
- c. x = 11 and y = 11
- d. x = -11 and y = 11

Answer: b. x = 10 and y = -10 ✓

Explanation: x remains 10, y becomes -10.

2. If m = 50, n = 5, then n % m is _____.

- a. 5
- b. 10
- c. 0
- d. 1

Answer: a. 5 ✓

Explanation: 5 % 50 = 5

3. “Change after action” is example of _____.

- a. Prefix operator
- b. Postfix operator
- c. Binary operator
- d. Assignment operator

Answer: b. Postfix operator ✓

4. Operators that contains one operand or expression _____.

- a. Unary operator
- b. Ternary operator
- c. Binary operator
- d. Null operator

Answer: a. Unary operator ✓

5. Which operator is used to initialize non-primitive data types?

- a. . (dot operator)
- b. Ternary operator
- c. New operator
- d. Relational

Answer: c. New operator ✓

6. What is result of $a += 9$?

- a. $++a$
- b. $a = a + 9$
- c. $a + 9$
- d. None

Answer: b. $a = a + 9$ ✓

7. If $a = 5$, $b = 4$, $c = 0$; find $c = a \% ++b$;

- a. 5.0
- b. 5
- c. 0
- d. None

Answer: c. 0 ✓

Explanation: $++b \rightarrow b = 5 \rightarrow 5 \% 5 = 0$

8. Output of $++a + ++a$; when $a = -1$

- a. -1
- b. 0
- c. 1
- d. None

Answer: c. 1 ✓

9. Correct Java expression for $p = a^2 + bc$

- a. $p = a * a + b * c$
- b. $p = axa + bxc$
- c. $p = a.a + b.c$
- d. None

Answer: a. $p = a * a + b * c$ ✓

10. Shorthand operator to decrease by 3

- a. $a++$
- b. $a -= 3$
- c. $a += 3$
- d. $a = 3$

Answer: b. $a -= 3$ ✓

11. Correct statement about print and println

- a. print prints newline
- b. println prints space
- c. print does not print newline
- d. println only prints integers

Answer: c. print does not print newline ✓

12. Output of code:

```
System.out.println("A");  
System.out.println("B");  
System.out.println("C");
```

- a. ABC
- b. A B C
- c. AB
- d. A, B, C

Answer: b. A B C (each on new line) ✓

13. Difference between prefix and postfix increment

- a. Prefix increments before use
- b. Postfix increments after use
- c. Both
- d. None

Answer: c. Both ✓

14. `int a = 7; int b = a++;` values?

- a. $a = 6, b = 7$
- b. $a = 7, b = 8$
- c. $a = 7, b = 7$
- d. $a = 8, b = 7$

Answer: d. $a = 8, b = 7$ ✓

15. Postfix decrement operator statement

- a. decreases by 2
- b. decreases before use
- c. decreases after use
- d. no change

Answer: c. decreases after use ✓

✓ B. Fill in the blanks

1. _____ are used to instruct compiler to execute operations.

Answer: Operators

2. _____ operator is used to access members of class.

Answer: dot operator

3. _____ operator is used to create objects.

Answer: new

4. Output of logical operator is in _____ form.

Answer: boolean

5. Counters increases by _____ and accumulator increases by _____.

Answer: 1, value

✓ C. Answer the following questions

1. Output:

- $r = m++ / ++n$
- $r = ++m + ++n - n++$

☞ Solve stepwise (student explanation needed)

2. Explain Arithmetic operators

Answer: Used for calculations (+, -, *, /, %)

3. Evaluate:

- $A = 3 + 2 \times 4 = 11$
- $B = 22 / 2 + 4 = 15$

4. Output:

- $a = ++a + ++a \rightarrow$ evaluated stepwise
- $i *= j + \% j \rightarrow$ apply precedence

5. Operators:

- a. $\sim \rightarrow$ Bitwise NOT
- b. $++ \rightarrow$ Increment
- c. $\&\& \rightarrow$ Logical AND

✓ D. Picture Study

1. > and < are _____ operators

- a. Logical
- b. Relational
- c. Assignment
- d. Conditional

Answer: b. Relational ✓

2. ! operator is _____

- a. Relational
- b. Not (Logical)
- c. And
- d. Assignment

Answer: b. Not (Logical) ✓

3. Output:

```
System.out.println(x == 7);
```

Answer: true (if $x = 7$) ✓

4. Ternary operator is also called _____

- a. Relational
- b. Arithmetic
- c. Conditional
- d. Logical

Answer: c. Conditional ✓

5. Operator checking condition is _____

- a. Relational
- b. Logical
- c. Conditional
- d. Arithmetic

Answer: a. Relational ✓

✓ E. Case Study

1. $a + b * c \rightarrow$ first operation?

- a. Addition
- b. Multiplication
- c. Addition first
- d. Simultaneous

Answer: b. Multiplication ✓

2. Operator with left-to-right associativity

- a. *
- b. =
- c. ?:
- d. ++

Answer: a. * ✓

3. $x = a - b + c$ evaluation

- a. c added first
- b. b subtracted then c added
- c. precedence only
- d. right to left

Answer: b. b subtracted then c added ✓

4. Right-to-left associativity operator

- a. +
- b. &&
- c. =
- d. -

Answer: c. = ✓

5. $a + b * c$ first executed

Answer: Multiplication ✓

✓ F. Assertion & Reasoning

1.

Assertion: OR evaluates both operands ✗

Reason: OR returns true if one is true ✓

Answer: d. A is false, R is true ✓

2.

Assertion: ?: has higher precedence than = ✓

Reason: Higher precedence executes first ✓

Answer: a. Both true and correct explanation ✓

Class 9
Computer Applications(2025-26)
Chapter: 5

Operators In Java
(Solutions are below the questions)

Unsolved Questions

A. Tick (✓) the correct answer

1. What values will be stored in x and y respectively after executing the following?

```
int x = -10;  
y = --x;
```

- a. x = -11, y = -11
 - b. x = -10, y = -10
 - c. x = 11, y = 11
 - d. x = -11, y = 11
-

2. If $m = 50$ and $n = 5$ then $n \% 2 = ?$

- a. 5
 - b. 10
 - c. 0
 - d. None of these
-

3. "Change after action" is the example of

- a. Prefix operator
 - b. Postfix operator
 - c. Binary operator
 - d. None of these
-

4. Operators that contain one operand or expression:

- a. Unary operator
 - b. Ternary operator
 - c. Binary operator
 - d. None of these
-

5. Which of the following operators is used to initialise all non-primitive data types?

- a. .(dot) operator
 - b. Ternary operator
 - c. New operator
 - d. Relational
-

6. What is the result of `a += 9;` statement?

- a. ++a
 - b. `a = a + 9`
 - c. `a + 9`
 - d. None of these
-

7. If `int a = 5, b = 4, c = 0;` what value is stored in `c`, when `c = a % ++b;`?

- a. 5.0
 - b. 5
 - c. 0
 - d. None of these
-

8. What will be the output of `++a+ ++a;` when `int a = -1`?

- a. -1
 - b. 0
 - c. 1
 - d. None of these
-

9. What is the correct Java expression for the statement $p = a^2 + bc$?

- a. `p = a*a + b*c;`
 - b. `p = axa + bxc;`
 - c. `p = a.a + b.c;`
 - d. None of these
-

10. The operators that deal with two operands are known as:

- a. Unary
- b. Binary

- c. Ternary
 - d. None of these
-

B. Fill in the blanks

1. Arrange these operators (<, ++, +, *) in order of higher precedence to lower precedence: _____
 2. Write the Java expression of a^2+b^2+2ab : _____
 3. Write the output of the expression $a++ + a \% a$, where $a = 5$: _____
 4. The output of Logical Operator is in _____ form.
 5. Counters increase by _____ and in accumulator, the increment value is _____ for each recurrence of the loop.
-

C. Short Answer Questions

1. What is the output of the following expressions if executed sequentially, if $m = 15$, $n = 5$:

- $r = m++ / ++n$
- $r = ++m + ++n - n++$

2. What are the Arithmetic operators?
-

3. Write the Java Expression of the following mathematical expressions:

- a. $A = (B + C)/2 * h$
 - b. $V = \pi * r^2 * h$
-

4. Give the output of the following expression

- $a = ++a + a - ++a - a$; when $a = 2$:
 - $i *= j++ \% j - k * 10$;
when $i = 2$, $j = 4$, $k = 3$.
-

5. Name the operators listed below:

- (i) <
 - (ii) &&
 - (iii) ++
 - (iv) ?:
-

6. Write the output of the following code:

```
char ch = 'F';
int m = ch;
m = m + 5;
System.out.println(m + "" + ch);
```

7. What is meant by precedence of operators?

8. Differentiate between the following:
a. Arithmetical operator and Logical operator
b. Logical AND and Logical OR

9. Define Postfix Increment Operator.

10. What will be the output of the following code?

```
int m = 2, n = 15;
for(int i = 1; i < 5; i++) {
    m++;
    n--;
}
System.out.println("m=" + m);
System.out.println("n=" + n);
```

◆◆◆ End of Worksheet ◆◆◆

Solutions – Operators (Class 9)

A. Tick (✓) the correct answer

1. int x = -10;
y = --x;
int x = -10; → so x = -10.

--x → pre-decrement, so x becomes -11 first, then the value -11 is assigned to y.

✓ Final values: x = -11, y = -11

Correct answer: a. x = -11, y = -11

2. If $m = 50$ and $n = 5 \rightarrow n \% 2 = 5 \% 2 = 1$.

But options are (5, 10, 0, None). Correct = **None of these** ✓

3. "Change after action" → **Postfix operator** ✓

4. One operand only → **Unary operator** ✓

5. To initialize objects (non-primitive) → **new operator** ✓

6. `a += 9;` means `a = a + 9.` ✓

7.

```
int a = 5, b = 4, c = 0;  
c = a % ++b;
```

Here `++b = 5`, so `c = 5 % 5 = 0`.

Answer: **0** ✓

8.

```
int a = -1;  
++a + ++a;
```

Step 1: `++a` → `a = 0`, gives 0.

Step 2: `++a` → `a = 1`, gives 1.

So total = `0 + 1 = 1` ✓

9. Correct Java expression for `p=a2+bcp = a^2 + bcp=a2+bc`:

`p = aa + bc;` ✓

10. Two operands → **Binary operator** ✓

B. Fill in the blanks

1. Operator precedence: $++$, $*$, $+$, $<$
 2. Java expression: $a*a + b*b + 2*a*b$
 3. Expression: $a += a++ + a \% a$ with $a=5 \rightarrow a = 5 + (5 + 0) \rightarrow 10$.
 4. Logical operator output is in **boolean** form.
 5. Counters increase by **1**, and in accumulator increment value is **variable (as per recurrence)**.
-

C. Short Answer Questions

1. If $m = 15, n = 5$:

- $r = m++ / ++n;$
 $m++ = 15, ++n = 6 \rightarrow r = 15 / 6 = 2$. (Then $m=16, n=6$).
 - $r = ++m + ++n - n++;$
 $m=17, n=7 \rightarrow r = 17 + 7 - 7 = 17$.
(After, $n=8$).
-

2. **Arithmetic operators:** $+$, $-$, $*$, $/$, $\%$
-

3. Java expressions:

- a. $A = (B + C) / 2 * h;$
 - b. $V = 3.14 * r*r * h;$
-

4. (i) If $a = 2$:

$a = ++a + a - ++a - a;$
 $= (3 + 3 - 4 - 4) = -2$

- (ii) $i *= j++ \% j - k * 10;$ when $i=2, j=4, k=3$

- First $j++ \% j = 4 \% 5 = 4$
 - So $\rightarrow 2 * (4 - 30) = 2 * -26 = -52$
-

- 5.

(i) $< \rightarrow$ Relational operator

(ii) $\&\& \rightarrow$ Logical AND

- (iii) ++ → Increment (Unary) operator
 - (iv) ?: → Ternary operator
-

6.

```
char ch = 'F';    // ASCII = 70
int m = ch;       // m = 70
m = m + 5;        // 75
System.out.println(m + " " + ch);
```

Output: 75F

7. **Precedence of operators:** The priority/order in which operators are evaluated in an expression.
-

8. Differences:

a. **Arithmetic:** Perform math (+, -, *, /, %).

Logical: Deal with Boolean (&&, ||, !).

b. **Logical AND (&&):** True only if both conditions true.

Logical OR (||): True if at least one condition true.

9. **Postfix Increment Operator:** Variable is incremented **after** its current value is used. Example:

```
int a = 5;
System.out.println(a++); // prints 5, then a becomes 6
```

10.

```
int m = 2, n = 15;
for(int i = 1; i < 5; i++) {
    m++;    // runs 4 times → m = 6
    n--;    // runs 4 times → n = 11
}
System.out.println("m=" + m); // m=6
System.out.println("n=" + n); // n=11
```